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II. Short Answers Questions

1. What is meant by periodic and non-periodic motion?. Give any two examples, for each motion.
2. What is meant by force constant of a spring?.
3. Define time period of simple harmonic motion.
4. Define frequency of simple harmonic motion.
5. What is an epoch?.
6. Write short notes on two springs connected in series.
7. Write short notes on two springs connected in parallel.
8. Write down the time period of simple pendulum.
9. State the laws of simple pendulum?.
10. Write down the equation of time period for linear harmonic oscillator.
11. What is meant by free oscillation?.
12. Explain damped oscillation. Give an example.
13. Define forced oscillation. Give an example.
14. What is meant by maintained oscillation?. Give an example.
15. Explain resonance. Give an example.

III. Long Answers Questions

1. What is meant by simple harmonic oscillation?. Give examples and explain why every simple harmonic motion is a periodic motion whereas the converse need not be true.
2. Describe Simple Harmonic Motion as a projection of uniform circular motion.
3. What is meant by angular harmonic oscillation?. Compute the time period of angular harmonic oscillation.
4. Write down the difference between simple harmonic motion and angular simple harmonic motion.
5. Discuss the simple pendulum in detail.
6. Explain the horizontal oscillations of a spring.
7. Describe the vertical oscillations of a spring.
8. Write short notes on the oscillations of liquid column in U-tube.
9. Discuss in detail the energy in simple harmonic motion.
10. Explain in detail the four different types of oscillations.

IV. Numerical Problems

1. Consider the Earth as a homogeneous sphere of radius R and a straight hole is bored in it through its centre. Show that a particle dropped into the hole will execute a simple harmonic motion such that its time period is

$$T=2\pi\sqrt{\frac{R}{g}}$$

2. Calculate the time period of the oscillation of a particle of mass m moving in the potential defined as

$$U(x)=\begin{cases}\frac{1}{2}kx^2, x<0\\ mgx, x>0\end{cases}$$

Answer: $\pi\sqrt{\frac{m}{k}}+2\sqrt{\frac{2E}{g^2m}}$, where E is the total energy of the particle.